

Internship Report

Frontend Web Developer

Wireframe to a fully responsive, accessible, interactive Single Page Application

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CLASS	3CSE24
BATCH	2024–2028
DEGREE	B.Tech CSE

```
<html>

  <body class="responsive">

    <navbar />

    <hero />

  </body>

</html>
```

Introduction

This report presents the work carried out during a five-week Summer Internship with YuvaIntern, in the role of Frontend Web Developer Intern — conducted remotely from July 22, 2026 to August 26, 2026, as part of the B.Tech CSE programme at IILM University, Greater Noida.

The internship progressed from a static wireframe conversion through responsive design, JavaScript interactivity, accessibility & performance optimization, to a simulated Single Page Application — each week building cumulatively on the last.

AT A GLANCE



5 Weeks

Duration



Remote

Mode of Internship



YuvaIntern

Organization (with NSDC)



Jul 22 – Aug 26, 2026

Internship Period

Organization Profile & Problem Statement



YuvaIntern

Operates in association with NSDC (National Skill Development Corporation) under the “Re-Imagine Future” initiative, offering virtual internship programs and job simulations to student interns. The program was led by Mr. Kounal Gupta, Founder, YuvaIntern.com.

The internship was conducted remotely, involving tasks and virtual job simulations in frontend web development, with compliance to organizational policy and confidentiality requirements.



The Problem

Modern websites are accessed across desktops, tablets and phones — each with different screen sizes and interaction methods. A page built without responsiveness, interactivity and accessibility in mind fails to deliver a consistent experience.

The task: take a given wireframe and evolve it into a fully responsive, interactive, accessible interface — then extend it into a simulated Single Page Application with working client-side navigation.

Project Objectives

01

Wireframe → Webpage

Study and convert a given wireframe into a static webpage using HTML5 and CSS3.

02

Responsive Design

Implement Flexbox, CSS Grid and media queries for desktop, tablet and mobile devices.

03

JavaScript Interactivity

Integrate DOM manipulation, event handling and form validation for a dynamic UX.

04

Accessibility & Performance

Apply semantic HTML, ARIA attributes, keyboard navigation and Lighthouse testing.

05

SPA Simulation

Develop dynamic content loading and routing using the Browser History API.

06

Hands-on Practice

Gain practical frontend development experience through weekly, task-based work.

Scope & Technologies Used

SCOPE OF THE PROJECT

- Week 1 — Wireframe converted into a responsive static webpage
- Week 2 — Responsive nav, images & cross-device layout refinement
- Week 3 — JavaScript interactivity: mobile menu & form validation
- Week 4 — Accessibility & performance: semantic HTML, ARIA, Lighthouse
- Week 5 — SPA simulation with client-side routing (History API)

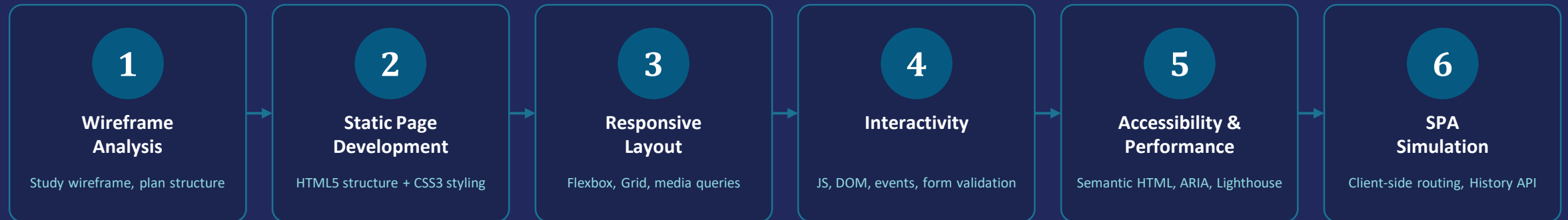
Limited strictly to HTML5, CSS3 and JavaScript — no backend, databases, or external frameworks/libraries.

TECHNOLOGIES & TOOLS

HTML5	CSS3	Flexbox
CSS Grid	Media Queries	JavaScript (DOM)
Event Listeners	ARIA Attributes	History API
Chrome DevTools	Lighthouse	

Frontend Development Workflow

Fig. 4.7.1 — Each stage builds on the output of the previous one



Methodology — Weeks 1 to 3

Week 1

Wireframe to Responsive Static Page

WHAT WAS DONE

Built semantic HTML5 + CSS3 markup from the wireframe using Flexbox, CSS Grid and media queries.

OUTCOME

Foundation in translating design into working, layout-driven code.

Week 2

Responsive Web Design

WHAT WAS DONE

Implemented a responsive navigation bar and images/elements that resize and reflow across breakpoints.

OUTCOME

Stronger skills in fully responsive interfaces and DevTools-based testing.

Week 3

JavaScript Interactivity

WHAT WAS DONE

Added a mobile nav menu and a contact form with client-side validation using event listeners and `preventDefault()`.

OUTCOME

Practical experience in DOM manipulation and event-driven programming.

Methodology — Weeks 4 & 5

Week 4

Performance & Accessibility

WHAT WAS DONE

Added skip-to-content links, visible focus states, ARIA labels and reduced-motion support; audited with Lighthouse.

100 / 100 / 100 / 100

Result verified through DevTools & Lighthouse testing

OUTCOME

Perfect Lighthouse scores across Performance, Accessibility, Best Practices and SEO.

Week 5

Single Page Application Simulation

WHAT WAS DONE

Built dynamic Home/About/Services/Contact sections switched via `history.pushState()` and the `popstate` event.

4 routes, 0 reloads

Result verified through DevTools & Lighthouse testing

OUTCOME

Understanding of client-side routing and SPA navigation without full page reloads.

Expected Outcomes



Ability to convert a wireframe into a working, responsive webpage using HTML5 and CSS3



Practical proficiency in responsive design with Flexbox, CSS Grid and media queries



Experience adding interactivity via JavaScript, DOM manipulation and event handling



Ability to build and validate web forms client-side, including email validation



Understanding of web accessibility principles: semantic HTML and ARIA attributes



Familiarity with Chrome DevTools and Lighthouse for testing and auditing



Understanding of Single Page Applications and client-side routing via History API



A structured, hands-on foundation in frontend web development

Certificate & Communication Proof

Chapter 4.10 · Appendix A



Internship Completion Certificate

Certificate of Experience issued by YuvaIntern, confirming successful completion of the internship as Virtual Data Exploration / Frontend Web Developer Intern.

Issued: August 26, 2026

Fig. A.1



Internship Offer Letter

Official offer letter from YuvaIntern documenting the internship role, duration (5 weeks, remote) and terms of engagement.

Issued: July 22, 2026

Fig. A.2

FIGURE A.1

Internship Completion Certificate



CERTIFICATE OF EXPERIENCE

Issued by YuvaIntern

This certifies that Ashutosh Shukla successfully completed the internship with YuvaIntern in the role of Virtual Data Exploration Intern, demonstrating consistent dedication, professionalism, and integrity throughout the program.

ISSUED ON	August 26, 2026
ISSUED BY	Kounal Gupta, Founder — YuvaIntern.com
AFFILIATION	YuvaIntern, in association with NSDC

FIGURE A.2

Internship Offer Letter



FORMAL OFFER

Frontend Web Developer Intern

Formal offer extended by YuvaIntern for the role of Frontend Web Developer Intern — responsible for creating and implementing user interfaces using HTML, CSS and JavaScript, and collaborating with the development team on virtual job simulations.

DATE OF ISSUE	July 22, 2026
DURATION	5 weeks (Jul 22 – Aug 26, 2026)
LOCATION	Remote
ISSUED BY	Kounal Gupta, Founder — YuvaIntern.com

Bibliography & References

- 1 YuvaIntern Internship Offer Letter — issued to Ashutosh Shukla, dated July 22, 2026
- 2 YuvaIntern Certificate of Experience — issued to Ashutosh Shukla, dated August 26, 2026
- 3 Internship Report Format — School of Computer Science and Engineering, IILM University
- 4 Mozilla Developer Network (MDN Web Docs) — HTML, CSS and JavaScript documentation
- 5 W3C Web Accessibility Initiative (WAI) — Accessibility guidelines
- 6 Google Chrome Developers — Lighthouse documentation

Thank You

Frontend Web Developer Internship — YuvaIntern × IILM University

Ashutosh Shukla

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